

PaperPal: An AI-Powered Platform for Collaborative Reading and Discussion

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Abstract & Motivation

Challenge: Collaborative academic reading lacks engaging, integrated tools. Current solutions are fragmented across platforms, leading to unfocused discussions and unequal participation.



Our Innovation

Advanced Features

Attention-Weighted Highlighting

Heuristic algorithm tracks highlighting frequency and recency to identify high-interest sections requiring AI assistance.

Contextual AI Support

Hover-activated assistance provides related content from current paper and external

First integrated platform combining real-time collaboration, AI-powered assistance, and structured learning workflows for academic paper reading

Impact: Transforms passive reading into active, collaborative learning through intelligent AI support and synchronized group engagement.

🎯 Core Solution

PaperPal integrates **six key components** into a unified collaborative reading experience:

🔗 Interactive PDF Viewer

Real-time collaborative annotation with synchronized highlighting

💬 Embedded AI Chat

Context-aware GPT-4 assistant integrated directly into reading flow

📊 Tunable Summarization

🗂️ Democratic Navigation

sources, enhancing concept connections.

🗨️ Threaded Discussions

Each highlight generates dedicated discussion threads with markdown formatting and conversation history tracking.

📈 Engagement Analytics

Real-time monitoring of user activity patterns with gentle intervention prompts for inactive participants.

🛡️ Reliability & Safety

Retrieval-Augmented Generation (RAG) grounds AI responses in uploaded paper content, reducing hallucination risks while maintaining conversational intelligence.

📈 Technical Innovation

🏗️ Architecture Highlights

Four adaptive summary styles: overview, concise, detailed, example-based

Section voting system for coordinated group reading progression



Smart Quizzes

Auto-generated comprehension checks with personalized feedback



Peer Support

Intelligent pairing of high and low performers for collaborative learning



AI-Guided Learning Flow

1

Understanding Phase

AI identifies and explains unfamiliar terms, building shared foundation

2

Exploration Phase

- **Real-time Sync:** Firebase-powered chat with instant message delivery
- **PDF Processing:** pdf.js extraction with percentage-based highlight alignment
- **AI Integration:** GPT-4 with adaptive feedback loops and user preference learning
- **Cross-platform:** React-PDF-viewer with custom annotation tools

React

Firebase

GPT-4

pdf.js

pdf-lib

RAG

6

Integrated Features

4

Learning Phases

4

Summary Styles

100%

Real-time Sync



Future Directions



Planned Evaluation

- Task-based usability studies

Deep analysis with AI monitoring for confusion and providing clarifications

3

Synthesis Phase

AI consolidates annotations and discussions into unified summary

4

Related Work Phase

AI recommends relevant papers based on key discussion points

- Comprehension assessment metrics
- Group interaction analysis
- Participation balance evaluation



Next Phase Features

- **Asynchronous Support:** Distributed team workflows
- **Enhanced Accessibility:** Screen reader compatibility, scalable text
- **Advanced Analytics:** Learning pattern insights
- **Multi-language Support:** Global research collaboration



System Interface

PaperPal Interface Overview

A) Dynamic Summary

Adjustable AI-generated summaries for each section

B) Voting & Quiz

Section voting and comprehension checks

C) AI Help & Chat

Live chat with immediate AI assistance

D) Collaborative Annotation

Real-time highlighting and commenting

E) Action Toolbar

Quick access to essential commands

F) User Profile Controls

Activity management and user presence



Quiz & Feedback System

AI-Driven Assessment Flow

Poor Performance Response:

- Detailed explanations for incorrect answers
- Simplified content clarifications
- AI assistant support options
- Peer pairing recommendations

Collaborative Discussion:

- Real-time chat during quiz
- Peer explanation opportunities
- AI-moderated clarification sessions

 Attention Heatmap**Collective Focus Visualization****High Activity Sections****Medium Activity****Low Activity**

AI identifies discussion hotspots and provides targeted assistance for high-attention areas

 Contact & Resources**Lead Author**

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